Gym API

Developer Documentation

API Version:	0.2.0
Base URL (local):	`http://127.0.0.1:8000`
Protocol:	JSON over HTTP
Authentication:	Bearer JWT (`Authorization: Bearer ...`)
Optional policy header:	`X-API-Key` (compute gating)
OpenAPI:	`/openapi.json`

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Chapter 1

Introduction

1.1 What this API provides

Gym API is a REST-style service for:

- logging workouts (workouts, exercise logs, sets with optional RIR) and weigh-ins;
- browsing a reference catalog (exercises, muscles, equipment, goals);
- computing user analytics (streaks, gaps, best set volume, estimated 1RM);
- generating training *signals* and *insights* from rolling windows and a state-machine classifier.

1.2 Design principles

This documentation aims for:

- **Clarity**: examples for success and error responses.
- **Predictability**: consistent schemas and status codes.
- **Safety**: strong authentication and clear authorization boundaries.
- **Practicality**: workflow-oriented guides and copy-pastable requests.

1.3 Quickstart

1.3.1 Run the API locally

```
uvicorn app.main:app --reload
```

1.3.2 Explore the API

- Swagger UI: `/docs`

- ReDoc: `/redoc`
- OpenAPI JSON: `/openapi.json`

1.4 Conventions

1.4.1 Dates

All dates use ISO-8601:

```
YYYY-MM-DD
```

1.4.2 Pagination

List endpoints typically accept:

```
?skip=0&limit=50
```

1.4.3 Authentication header

Authenticated endpoints require:

```
Authorization: Bearer <access_token>
```

1.4.4 Optional compute policy header

Some compute-heavy endpoints accept:

```
X-API-Key: <key>
```

1.4.5 Error format

FastAPI-style errors:

```
{ "detail": "..." }
```

Validation errors (422) include locations:

```
{
  "detail": [
    {
      "loc": ["body", "password"],
      "msg": "String should have at least 8 characters",
      "type": "string_too_short"
    }
  ]
```

```
}
```

1.5 Versioning

The API exposes `info.version = 0.2.0` in OpenAPI. Backwards-incompatible changes should bump the major version.

Chapter 2

Architecture

2.1 High-level request flow

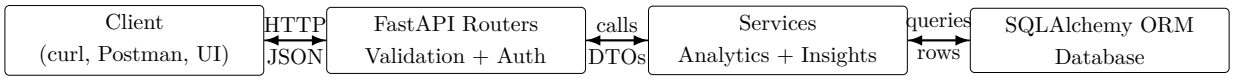

Figure 2.1: Request path from client to database and back

2.2 Data model (ER diagram)

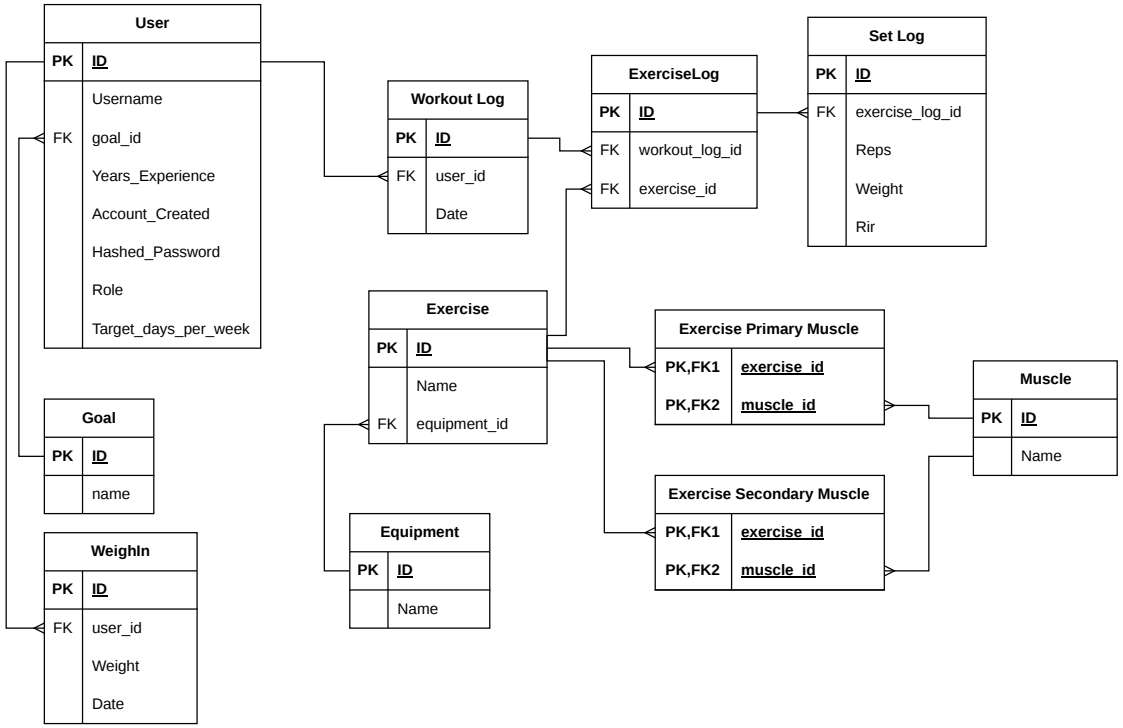

Figure 2.2: Entity Relationship Diagram of the Workout Tracking System

Chapter 3

Security

3.1 Authentication: JWT Bearer

Protected endpoints require:

```
Authorization: Bearer <access_token>
```

401 example (missing token)

```
{ "detail": "Not authenticated" }
```

3.2 Authorization: roles

The API supports:

- **user**: operates on their own data via `/me/...`.
- **admin**: can manage all users and any user-id scoped resources.

403 example (role not permitted)

```
{ "detail": "Admin role required" }
```

3.3 Why `/me` endpoints matter

User-scoped routes (`/me/...`) derive identity from the JWT, preventing a user from supplying a different `user_id` to view or mutate another user's data.

3.4 Optional compute gating: `X-API-Key`

Some compute-heavy endpoints accept `X-API-Key`. This is distinct from user authentication:

- JWT proves *who* is calling.
- API key can be used as an additional policy control for expensive endpoints.

403 example (API key policy)

```
{ "detail": "X-API-Key required for this endpoint" }
```

3.5 Recommended operational hardening

- Serve over HTTPS in production.
- Store secrets (JWT signing key, API key) in environment variables.
- Add request logging and rate limiting at a reverse proxy (e.g., Nginx).

Chapter 4

Common Workflows

4.1 Workflow 1: create an account and complete profile

4.1.1 Register

POST `/auth/register`

Request

```
{
  "username": "cayden01",
  "password": "StrongPass123"
}
```

Response (201)

```
{
  "access_token": "<jwt>",
  "token_type": "bearer"
}
```

4.1.2 Login

POST `/auth/login`

Request

```
{
  "username": "cayden01",
  "password": "StrongPass123"
}
```

Response (200)

```
{
  "access_token": "<jwt>",
  "token_type": "bearer"
}
```

Error (401)

```
{ "detail": "Invalid credentials" }
```

4.1.3 Set goal and training target

PATCH `/me/`

Request

```
{
  "goal_id": 2,
  "target_days_per_week": 4,
  "years_experience": 3
}
```

Response (200)

```
{
  "id": 1,
  "username": "cayden01",
  "years_experience": 3,
  "goal_id": 2,
  "target_days_per_week": 4,
  "account_created": "2026-03-05",
  "role": "user"
}
```

4.2 Workflow 2: log a workout (workout -> exercise logs -> sets)

4.2.1 Create workout

POST `/me/workouts`

Request

```
{ "date": "2026-03-05" }
```

Response (201)

```
{ "id": 10, "user_id": 1, "date": "2026-03-05" }
```

4.2.2 Attach an exercise

POST `/me/workouts/{workout_id}/exercise-logs`

Request

```
{ "exercise_id": 101 }
```

Response (201)

```
{
  "id": 55,
  "workout_id": 10,
  "exercise": {
    "id": 101,
    "name": "Barbell Bench Press",
    "equipment": { "id": 1, "name": "barbell" },
    "primary_muscles": [{ "id": 7, "name": "chest" }],
    "secondary_muscles": [{ "id": 12, "name": "triceps" }]
  },
  "sets": []
}
```

4.2.3 Log sets (RIR optional)

POST `/me/exercise-logs/{exercise_log_id}/sets`

Request

```
{ "reps": 8, "weight": 80.0, "rir": 2 }
```

Response (201)

```
{ "id": 901, "reps": 8, "weight": 80.0, "rir": 2 }
```

4.3 Workflow 3: insights and signals

4.3.1 Fetch signals

GET `/me/signals?as_of=2026-03-05`

Response (200)


```
{
  "user_id": 1,
  "signals": {
    "date_from": "2026-02-06",
    "date_to": "2026-03-05",
    "session_load_total": 1420.0,
    "acute_load_7d": 1420.0,
    "chronic_load_28d": 5150.0,
    "acwr": 1.10,
    "fatigue_trend_14d": 0.18,
    "workouts_last_7d": 4,
    "workouts_last_30d": 16,
    "adherence_ratio_30d": 1.00,
    "target_days_per_week": 4,
    "avg_rir_last_7d": 2.4,
    "hard_sets_rate_7d": 0.22,
    "prs_last_21d": 2,
    "last_pr_date": "2026-03-03",
    "goal_name": "Hypertrophy"
  }
}
```


4.3.2 Fetch insights

GET `/me/insights?as_of=2026-03-05`

Response (200)


```
{
  "user_id": 1,
  "signals": { "...": "see /me/signals" },
  "state": {
    "id": "on_track",
    "confidence": 0.60,
    "reasons": ["No major risk signals detected."]
  },
  "cards": [
    {
      "id": "on_track",
      "title": "You're on track",
      "severity": "success",
      "metrics": { "acwr": 1.10, "prs_last_21d": 2 },
      "evidence": ["Workload and performance look stable."],
      "actions": [
        {
          "type": "micro_progression",
```

```
          "label": "Aim for a small progression next session (+1 rep or
    +1.25-2.5kg)",
          "target": null,
          "params": {}
        }
      ]
    }
  ]
}
```


Potential error (403)


```
{ "detail": "X-API-Key required for this endpoint" }
```

Chapter 5

API Reference

5.1 Health

GET `/health`

Response (200)

```
{ "status": "ok" }
```

5.2 Authentication

5.2.1 Register

POST `/auth/register`

Success (201)

```
{ "access_token": "<jwt>", "token_type": "bearer" }
```

Error (422)

```
{
  "detail": [
    { "loc": ["body", "password"], "msg": "String should have at least 8
    characters", "type": "string_too_short" }
  ]
}
```

5.2.2 Login

POST `/auth/login`

Success (200)

```
{ "access_token": "<jwt>", "token_type": "bearer" }
```

Error (401)

```
{ "detail": "Invalid credentials" }
```

5.2.3 Who am I

GET `/auth/me`

Success (200)

```
{ "id": 1, "username": "cayden01", "role": "user" }
```

Error (401)

```
{ "detail": "Not authenticated" }
```

5.3 Me (user-scoped)

5.3.1 Profile

GET `/me/`

Success (200)

```
{
  "id": 1,
  "username": "cayden01",
  "years_experience": 3,
  "goal_id": 2,
  "target_days_per_week": 4,
  "account_created": "2026-03-05",
  "role": "user"
}
```

PATCH `/me/`

Request

```
{ "target_days_per_week": 5 }
```

Success (200)

```
{
  "id": 1,
  "username": "cayden01",
  "years_experience": 3,
  "goal_id": 2,
  "target_days_per_week": 5,
  "account_created": "2026-03-05",
  "role": "user"
}
```

5.3.2 Workouts

GET `/me/workouts?skip=0&limit=50`

Success (200)

```
[
  { "id": 10, "user_id": 1, "date": "2026-03-05" },
  { "id": 9,  "user_id": 1, "date": "2026-03-03" }
]
```

POST `/me/workouts`

Request

```
{ "date": "2026-03-06" }
```

Success (201)

```
{ "id": 11, "user_id": 1, "date": "2026-03-06" }
```

Error (422)

```
{
  "detail": [
    { "loc": ["body", "date"], "msg": "Input should be a valid date", "type":
    "date_from_datetime_parsing" }
  ]
}
```

GET `/me/workouts/{workout_id}`

Success (200)

```
{ "id": 11, "user_id": 1, "date": "2026-03-06" }
```

Error (404)

```
{ "detail": "Workout not found" }
```

PATCH `/me/workouts/{workout_id}`

Request

```
{ "date": "2026-03-07" }
```

Success (200)

```
{ "id": 11, "user_id": 1, "date": "2026-03-07" }
```

DELETE `/me/workouts/{workout_id}`

Success (204)

```
(no body)
```

5.3.3 Exercise logs

GET `/me/workouts/{workout_id}/exercise-logs?skip=0&limit=50`

Success (200)

```
[
  {
    "id": 55,
    "workout_id": 10,
    "exercise": {
      "id": 101,
      "name": "Barbell Bench Press",
      "equipment": { "id": 1, "name": "barbell" },
      "primary_muscles": [{ "id": 7, "name": "chest" }],
      "secondary_muscles": [{ "id": 12, "name": "triceps" }]
    },
    "sets": [
      { "id": 901, "reps": 8, "weight": 80.0, "rir": 2 }
    ]
  }
]
```

POST `/me/workouts/{workout_id}/exercise-logs`

Request

```
{ "exercise_id": 101 }
```

Success (201)

```
{
  "id": 56,
  "workout_id": 10,
  "exercise": {
    "id": 101,
    "name": "Barbell Bench Press",
    "equipment": { "id": 1, "name": "barbell" },
    "primary_muscles": [{ "id": 7, "name": "chest" }],
    "secondary_muscles": [{ "id": 12, "name": "triceps" }]
  },
  "sets": []
}
```

Error (404)

```
{ "detail": "Workout not found" }
```

GET `/me/exercise-logs/{exercise_log_id}`

Success (200)

```
{
  "id": 56,
  "workout_id": 10,
  "exercise": {
    "id": 101,
    "name": "Barbell Bench Press",
    "equipment": { "id": 1, "name": "barbell" },
    "primary_muscles": [{ "id": 7, "name": "chest" }],
    "secondary_muscles": [{ "id": 12, "name": "triceps" }]
  },
  "sets": []
}
```

PATCH `/me/exercise-logs/{exercise_log_id}`

Request

```
{ "exercise_id": 205 }
```

Success (200)

```
{
  "id": 56,
  "workout_id": 10,
  "exercise": {
    "id": 205,
    "name": "Lat Pulldown",
    "equipment": { "id": 3, "name": "machine" },
    "primary_muscles": [{ "id": 9, "name": "back" }],
    "secondary_muscles": [{ "id": 13, "name": "biceps" }]
  },
  "sets": []
}
```

Error (404)

```
{ "detail": "Exercise log not found" }
```

DELETE `/me/exercise-logs/{exercise_log_id}`

Success (204)

```
(no body)
```

5.3.4 Sets

GET `/me/exercise-logs/{exercise_log_id}/sets?skip=0&limit=50`

Success (200)

```
[
  { "id": 901, "reps": 8, "weight": 80.0, "rir": 2 },
  { "id": 902, "reps": 8, "weight": 80.0, "rir": 2 }
]
```

POST `/me/exercise-logs/{exercise_log_id}/sets`

Request

```
{ "reps": 10, "weight": 60.0, "rir": null }
```

Success (201)

```
{ "id": 903, "reps": 10, "weight": 60.0, "rir": null }
```

Error (404)

```
{ "detail": "Exercise log not found" }
```

GET `/me/sets/{set_id}`

Success (200)

```
{ "id": 903, "reps": 10, "weight": 60.0, "rir": null }
```

PATCH `/me/sets/{set_id}`

Request

```
{ "weight": 62.5, "rir": 1 }
```

Success (200)

```
{ "id": 903, "reps": 10, "weight": 62.5, "rir": 1 }
```

DELETE `/me/sets/{set_id}`

Success (204)

```
(no body)
```

5.3.5 Weigh-ins

GET `/me/weigh-ins?limit=5`

Success (200)

```
{
  "user_id": 1,
  "count": 2,
  "weigh_ins": [
    { "id": 1, "user_id": 1, "weight": 76.7, "date": "2026-02-20" },
    { "id": 2, "user_id": 1, "weight": 76.4, "date": "2026-03-05" }
  ]
}
```

POST `/me/weigh-ins`

Request

```
{ "weight": 76.4, "date": "2026-03-05" }
```

Success (201)

```
{ "id": 3, "user_id": 1, "weight": 76.4, "date": "2026-03-05" }
```

GET `/me/weigh-ins/{weigh_in_id}`

Success (200)

```
{ "id": 3, "user_id": 1, "weight": 76.4, "date": "2026-03-05" }
```

PATCH `/me/weigh-ins/{weigh_in_id}`

Request

```
{ "weight": 76.2 }
```

Success (200)

```
{ "id": 3, "user_id": 1, "weight": 76.2, "date": "2026-03-05" }
```

DELETE `/me/weigh-ins/{weigh_in_id}`

Success (204)

```
(no body)
```

5.3.6 Analytics

GET `/me/analytics/weekly-streak?weekly_goal=3`

Success (200)

```
{ "user_id": 1, "weekly_goal": 3, "current_weekly_streak": 4 }
```

GET `/me/analytics/workouts-last-30-days`

Success (200)

```
{ "user_id": 1, "days": 30, "workouts_count": 16 }
```

GET `/me/analytics/last-week-missed`

Success (200)

```
{ "user_id": 1, "last_gap_start_date": null, "last_gap_end_date": null }
```

GET `/me/analytics/exercises/max-set-volume?exercise_id=101`

Success (200)

```
[
  {
    "exercise_id": 101,
    "exercise_name": "Barbell Bench Press",
    "max_set_volume": 640.0,
    "reps": 8,
    "weight": 80.0,
    "date": "2026-03-05"
  }
]
```

GET `/me/analytics/exercises/best-1rm?formula=epley&max_reps=12`

Success (200)

```
[
  {
    "exercise_id": 101,
    "exercise_name": "Barbell Bench Press",
    "formula": "epley",
    "best_e1rm": 101.3
  }
]
```

5.3.7 Insights

GET `/me/signals?as_of=2026-03-05`

GET `/me/insights?as_of=2026-03-05`

Note: These endpoints may also accept `X-API-Key` depending on configured policy.

5.4 Admin endpoints

Note: Admin endpoints are intended for global management and demonstration of privileged access.

5.4.1 Users

POST `/users/`

Request

```
{
  "username": "alice01",
  "password": "StrongPass123",
  "role": "user",
  "years_experience": 1,
  "goal_id": 4,
  "target_days_per_week": 5
}
```

Success (201)

```
{
  "id": 2,
  "username": "alice01",
  "years_experience": 1,
  "goal_id": 4,
  "target_days_per_week": 5,
  "account_created": "2026-03-05",
  "role": "user"
}
```

Error (403)

```
{ "detail": "Admin role required" }
```

GET `/users/?skip=0&limit=50`

Success (200)

```
[
  { "id": 1, "username": "cayden01", "years_experience": 3, "goal_id": 2,
    "target_days_per_week": 4, "account_created": "2026-03-05", "role": "user" },
  { "id": 2, "username": "alice01", "years_experience": 1, "goal_id": 4,
    "target_days_per_week": 5, "account_created": "2026-03-05", "role": "user" }
]
```

GET `/users/{user_id}`

Success (200)

```
{ "id": 1, "username": "cayden01", "years_experience": 3, "goal_id": 2,
    "target_days_per_week": 4, "account_created": "2026-03-05", "role": "user" }
```

PATCH `/users/{user_id}`

Request

```
{ "role": "admin" }
```

Success (200)

```
{ "id": 1, "username": "cayden01", "years_experience": 3, "goal_id": 2,
    "target_days_per_week": 4, "account_created": "2026-03-05", "role": "admin" }
```

DELETE `/users/{user_id}`

Success (204)

```
(no body)
```

5.5 Catalog (read-only reference data)

GET `/catalog/exercises?skip=0&limit=50&q=bench`

Success (200)

```
[
  {
    "id": 101,
    "name": "Barbell Bench Press",
    "equipment": { "id": 1, "name": "barbell" },
    "primary_muscles": [{ "id": 7, "name": "chest" }],
    "secondary_muscles": [{ "id": 12, "name": "triceps" }]
  }
]
```

GET `/catalog/exercises/{exercise_id}`

Error (404)

```
{ "detail": "Exercise not found" }
```

GET `/catalog/muscles`

GET `/catalog/equipment`

GET `/catalog/goals/`

Chapter 6

Operational Guide

6.1 Seeding and environments

The project includes seed scripts for repopulating reference and sample data:

- `seed_goals.py`
- `seed_exercises.py` (plus equipment and muscles)
- `seed_users.py`
- `seed_workouts.py`
- `seed_weigh_ins.py`

6.2 Suggested dependencies

```
fastapi
uvicorn[standard]
sqlalchemy
pydantic
python-jose[cryptography]
passlib[bcrypt]
python-multipart
```